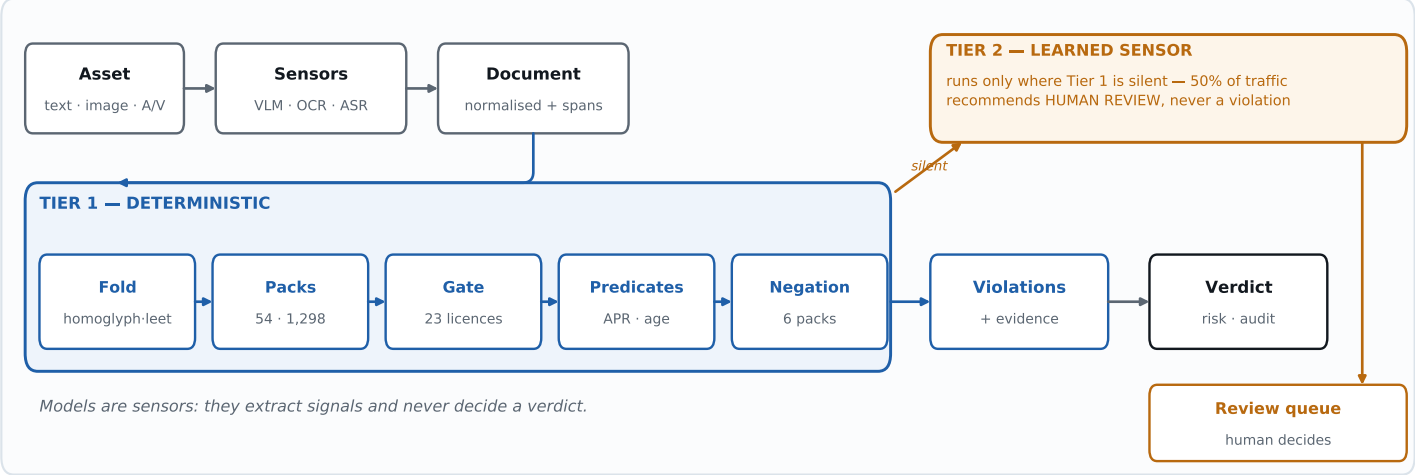


ARCHITECTURE



THE THREE CONFIGURATIONS

**DETERMINISTIC**

Guarded licensing, not classification.

$$V(d, p) = T(d, p) \wedge \neg E(d) \wedge C(d)$$
$$\wedge \neg \exists c \in Q(p) : L_c(W(d))$$

monotone in triggers, antitone in licences

825 terms · 287 phrases · 186 regex  
23 licence classes · 146 patterns  
window  $W = 240$  chars around the trigger  
36 packs licensable · 18 absolute

Risk  $R = \min(100, \sum_i \pi(s_i) \mu(c_i))$   
 $\pi$ : 100/50/20/5  $\mu$ : 1.0/0.7/0.3

**P 0.826 R 0.593 F1 0.690**

specificity 0.704 · ~5 ms/row

**LANGUAGE MODEL**

Linear, inspectable, reproducible.

$$x = [\hat{n}(d) ; \phi(d)] \in \mathbb{R}^{20010}$$
$$p = \sigma(w^T x + b)$$

$\hat{n}$ : L2-normalised 1-2gram + char-4gram  
 $\phi$ : 10 structural flags

$$J = -\sum_i \alpha_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$$
$$+ \frac{\lambda}{2} \|w\|^2$$

$w^{(0)} = 0$  · full batch  $\eta_t = \eta / (1 + t/100)$  · 400 it  
convex  $\Rightarrow$  path-independent  $\Rightarrow$  bit-identical weights

242 KB .npz · numpy only · no torch · offline

**P 0.828 R 0.869 F1 0.848**

specificity 0.571 · dev AUC 0.837

**BOTH (CASCADE)**

Tier 2 asked only where Tier 1 is silent.

$$d \vee (s \wedge \neg d) \equiv d \vee s$$

the same predicate — a cascade earns its name on cost, not on the decision

$$E[T] = \sum_i \left( \prod_{j < i} e_j \right) t_i$$

escalation rate  $e$  dominates, not tier speed

Where Tier 1 is silent (339 of 684 rows):  
recovers 86% of the violations it missed  
false-alarms on 38% of the clean copy

confirmed ( $d \wedge s$ ): P 0.883 · spec 0.837

**P 0.799 R 0.942 F2 0.909**

specificity 0.438 · recall +58%

THE MATHEMATICS THAT DECIDES WHAT THESE NUMBERS MEAN

**1 · Why this is not a classifier**

$I(\text{words}; y) \approx 0$

85% of compliant minimal pairs carry the same trigger as the violation they pair with, by construction. Vocabulary cannot separate them; the licence gate can. The information is in the structure, not in the words.

**2 · Calibration is empirical Bayes**

$$\hat{c} = \frac{k + m\pi}{n + m}, \quad m = 8$$

Beta-Binomial posterior mean. Every regex shipped at 0.88; measured, they range 0.19 to 0.64.

**3 · Variance is clustered**

$$SE = \sqrt{p(1-p)/G}, \quad G = 128$$

Rows inside one enforcement case are not independent — they restate one claim. Effective  $n$  is 128 cases, not 347 rows.

$P = 0.824 \pm 0.066$ , not  $\pm 0.040$

design effect 1.65 × — compare engines with a cluster bootstrap over cases, or manufacture significance from correlated rows.

**4 · The loss is asymmetric**

$$L = C_{FN} \cdot FN + C_{FP} \cdot FP, \quad C_{FN} > C_{FP}$$

A missed violation costs more than a flagged ad, so report  $F_2 = \frac{5PR}{4P+R}$  rather than  $F_1$ .

**5 · Evidence is double-counted**

now:  $R = \sum_i \pi_i \mu_i$  (assumes independence)

"cure for cancer" — three words, one span — fires 8 packs and scores 100 out of 100. That measures how densely the pack library covers a phrase, not how bad the ad is.

**fix:**  $P = 1 - \prod_i (1 - p_i)$  over distinct spans

bounded, order-independent, and immune to adding redundant packs.

**6 · The split unit is the case**

1,589 rows  $\leftarrow$  827 cases  $\Rightarrow$  split by case

578 / 121 / 128 groups. 13 north-star cases are locked to test, so a harvested sibling can never leak into training.

WHAT THE NUMBERS DO NOT SAY

**The model did not learn the rule.**

Recall by source on the held-out split. \* in-sample: the miner tokenised those rows straight into pack terms.

| source                          | det    | model        |
|---------------------------------|--------|--------------|
| harvested — enforcement wording | 0.490  | <b>0.908</b> |
| precedents — expert, north star | 0.932* | <b>0.864</b> |
| seed — expert, third register   | 0.732  | <b>0.772</b> |
| compliant pairs — specificity   | —      | <b>0.564</b> |

0.564 on minimal pairs is a coin flip. It learned that enforcement-quoted copy reads differently from a rewrite — not the rule. Its own explain() agrees: top features on escalated copy are "c: the" and "w:the".

HELD-OUT RESULTS

684 rows · 128 enforcement cases · test split

| configuration              | P            | R            | F1           | F2           | spec         |
|----------------------------|--------------|--------------|--------------|--------------|--------------|
| deterministic              | 0.826        | 0.593        | 0.690        | 0.628        | 0.704        |
| language model             | 0.828        | 0.869        | 0.848        | 0.860        | 0.571        |
| cascade ( $d \vee s$ )     | <b>0.799</b> | <b>0.942</b> | <b>0.865</b> | <b>0.909</b> | <b>0.438</b> |
| confirmed ( $d \wedge s$ ) | 0.883        | 0.520        | 0.654        | 0.566        | 0.837        |

Recall 0.593 $\rightarrow$ 0.942 and  $F_2$  0.628 $\rightarrow$ 0.909 are real.  
Specificity 0.704 $\rightarrow$ 0.438 is the price, and in a review queue that price is reviewer time — so the learned tier ships behind a flag, off.